Microbial quality of raw aquacultured fish fillets procured from Internet and local retail markets.

The microbial quality of raw fillets of aquacultured catfish, salmon, tilapia, and trout was evaluated. A total of 272 fillets from nine local and nine Internet retail markets were tested. Mean values were 5.7 log CFU/g for total aerobic mesophiles, 6.3 log CFU/g for psychrotrophs, and 1.9 log most probable number (MPN) per gram for coliforms. Differences in these microbial levels between the two kinds of markets and among the four types of fish were not significant ($P > 0.05$), except that Internet trout fillets had about 0.8-log higher aerobic mesophiles than did trout fillets purchased locally. Although Escherichia coli was detected in 1.4, 1.5, and 5.9% of trout, salmon, and tilapia, respectively, no sample had > or = 1.0 log MPN/g. However, E. coli was found in 13.2% of catfish, with an average of 1.7 log MPN/g. About 27% of all fillets had Listeria spp., and a positive correlation between the prevalence of Listeria spp. and Listeria monocytogenes was observed. Internet fillets had a higher prevalence of both Listeria spp. and L. monocytogenes than did those fillets purchased locally. L. monocytogenes was present in 23.5% of catfish but in only 5.7, 10.3, and 10.6% of trout, tilapia, and salmon, respectively. Salmonella and E. coli O157 were not found in any sample. A follow-up investigation using catfish operation as a model revealed that gut waste exposed during evisceration is a potential source of coliforms and Listeria spp.